Virtual H&E Staining Capstone Project

Detailed Progress Report From Early Experiments to Final Model

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How to Read This Report

This report is written for a first-year college student who may not already know deep learning, histology, or medical image registration. The goal is to explain the project in simple language while still giving enough technical detail to defend the work in a review or presentation.

The project goal was to build a system that takes an unstained tissue microscopy image and predicts how the same tissue would look after H&E staining. H&E means Hematoxylin and Eosin. In normal pathology, Hematoxylin stains cell nuclei purple-blue and Eosin stains cytoplasm and extracellular tissue pink. Here, instead of physically staining the sample, we train a model to create a virtual H&E image from the unstained image.

The final best system is not one single trick. It is the result of many decisions: better image registration, better data filtering, stronger neural network backbones, simpler losses, clean validation splits, test-time augmentation, and a final ensemble. This report explains why each decision was made and why some earlier ideas were stopped.

Executive Summary

The final project pipeline is:

```
 Unstained image pair data
-> CLAHE-driven TV-L1 non-rigid registration
-> content-quality top-1000 training selection
-> v20/v22A/v21B model family
-> 4-flip test-time augmentation
-> weighted final ensemble
-> desktop/CLI staining app
```

The final recommended model mode is the balanced ensemble:

```
 0.20 * v20_fixed TTA4
0.60 * v22A content-quality TTA4
0.20 * v21B Hibou-B content-quality TTA4
```

On the final fixed CLAHE validation set, this scored:

Metric	Final Balanced Ensemble
SSIM	0.7838
PSNR	25.16 dB
PCC	0.8794

These numbers come from the project file `logs/final_v21b_v22a_eval_summary.txt` and are also recorded in `best_model_manifest.json` [P1, P2].

The best single model on the same CLAHE fixed validation set was v22A TTA4, with:

Metric	v22A TTA4
SSIM	0.7807
PSNR	25.16 dB
PCC	0.8782

The final ensemble only improved slightly over v22A alone, but the improvement was consistent enough to keep it as the highest-quality internal mode. For a simpler public app, v22A-only or v20-only remains easier to distribute because v21B depends on the Hibou-B model files.

The most important lesson from the project is simple:

> In virtual staining, better alignment and cleaner data mattered more than adding complicated GAN losses.

The early models failed mostly because the unstained and stained images were not perfectly aligned. Later models improved because registration and data quality improved. The best final result came from combining a strong ConvNeXt-based model with better registered CLAHE data and a small ensemble.

Beginner-Friendly Glossary

Histology means studying tissue under a microscope.

H&E staining is a common staining method used in pathology. It colors nuclei and tissue structures so doctors can inspect them.

Virtual staining means using a computer model to predict a stained-looking image from an unstained image.

Registration means aligning two images of the same tissue so the same cell appears in the same place in both images.

Non-rigid registration means the alignment can bend and stretch locally, not just shift the whole image left or right.

Optical flow estimates how each pixel should move from one image to another. TV-L1 optical flow is a robust optical-flow method that uses total variation smoothing and L1 brightness error [R1].

CLAHE means Contrast Limited Adaptive Histogram Equalization. It improves local contrast in an image without letting noise become too strong [R9].

U-Net is a neural network architecture widely used in biomedical image segmentation and image-to-image problems because it keeps both high-level context and fine spatial details [R3].

Pix2Pix is a conditional GAN framework for image-to-image translation. It learns a mapping from an input image to an output image [R4].

GAN means Generative Adversarial Network. It trains a generator and a discriminator against each other. GANs can make images look sharper, but they can also become unstable.

L1 loss measures average absolute pixel error. It is simple and stable.

Perceptual loss compares deep features instead of raw pixels. VGG-19 features are often used for this because VGG networks learned useful visual features on ImageNet [R8].

SSIM means Structural Similarity Index. It measures how similar two images are in structure, contrast, and brightness. It is often better than pure pixel error for image quality [R2].

PSNR means Peak Signal-to-Noise Ratio. Higher PSNR usually means lower pixel-level error.

PCC means Pearson Correlation Coefficient. It measures how strongly two images vary together.

TTA means Test-Time Augmentation. In this project, TTA4 means predicting the image four times using flips, then flipping predictions back and averaging them.

Ensemble means combining predictions from multiple models. Ensembles often improve reliability because different models make slightly different errors.

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1. Problem Statement

The project tries to answer this question:

> Can we generate a realistic H&E-stained image from an unstained tissue image using deep learning?

This is useful because physical staining takes time, chemicals, and tissue handling. A virtual staining model could help create a fast preview, reduce repeated staining, or support computational pathology workflows. It does not replace clinical validation, but it can be a strong research step.

The task is difficult because the input and output images are not just different colors. During staining, the tissue can move, stretch, compress, or slightly deform. If the input unstained image and target stained image are not aligned, a model gets contradictory training signals. For example, the model may see a nucleus at location `(x, y)` in the input but the target nucleus is shifted 20 pixels away. The model is then punished even if it predicted the correct biological structure in the wrong position.

This is why registration became the foundation of the project. Without registration, the model learns blur. With registration, the model can learn staining.

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2. What Data the Project Used

The project used paired microscopy patches. Each pair contains:

1. an unstained tissue image, and
2. the corresponding H&E stained image.

The pairs are related, but they are not naturally pixel-perfect. Tissue processing and microscope repositioning cause mismatch.

The project eventually worked with 8,885 registered pairs after running full registration pipelines. Different training subsets were created from these pairs:

Data subset	Purpose
All registered pairs	Useful for global statistics, but includes low-quality alignments
Top-1000 by registration score	Higher-quality aligned training subset
Content-quality top-1000	Final stronger subset balancing registration quality and tissue information
Fixed same-prefix validation set	Fair final comparison across models

The final model claim uses a fixed validation protocol instead of relying only on internal training validation. This matters because some earlier results were later found to have overlapping train/validation prefixes. Those earlier results were useful for learning, but they should not be presented as clean final validation.

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3. Why Registration Was the First Big Challenge

Imagine tracing a drawing on butter paper. If the paper moves while you are tracing, your drawing looks wrong even if your hand is good. The same thing happens in virtual staining. The model can only learn well if the input and target are aligned.

At the beginning, the model was trained on unregistered data. The result was poor:

Stage	SSIM
v1-v7, unregistered Pix2Pix	about 0.26

The images looked like blurry pink-purple regions instead of clean cell structures. This was not mainly a model problem. It was a data-alignment problem.

The team tested multiple registration methods:

Method	What it can do	Why it failed or worked
Phase correlation	Global shift only	Too simple for tissue stretching
ORB feature matching	Feature-based alignment	Failed because nuclei look repetitive and cross-modal features differ
SyN/windowed SyN	Advanced deformable registration	More complex and not clearly better for this pipeline
TV-L1 optical flow	Dense non-rigid alignment	Worked best historically
CLAHE-driven TV-L1	TV-L1 with better contrast drivers	Best final registration path

TV-L1 optical flow was chosen because it estimates a motion vector for each pixel, so it can model local tissue deformation. This matches the biological reality: tissue does not only shift; it bends and stretches. The method is based on a variational optical-flow formulation using total variation regularization and L1 data fidelity [R1].

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4. Why CLAHE Improved Registration

The original TV-L1 registration used grayscale images to estimate the optical flow. That helped a lot, but grayscale unstained and grayscale H&E images still do not always have the same contrast. Some structures are clearer in one domain than the other.

CLAHE was introduced as a preprocessing step only for estimating the registration flow. CLAHE increases local contrast in a controlled way. It helps the optical-flow algorithm see tissue edges and nuclei-like structures more clearly. It was not used as an inference-time image enhancement step in the final app. In other words:

```
 CLAHE is used to help align image pairs during data preparation.
CLAHE is not applied to a single user image inside the app.
```

The improvement was large:

Registration Dataset	All-Pair Mean SSIM
Old gray TV-L1	0.4134
CLAHE TV-L1	0.5045
Mean gain	+0.0911

For the top-1000 registered pairs:

Dataset	Top-1000 Mean SSIM
Old gray TV-L1	0.6094
CLAHE TV-L1	0.6423

This was one of the most important discoveries in the project. It showed that the model was limited not only by architecture, but by label quality. Better aligned target images gave the learning system a clearer signal.

5. Metrics Used in the Project

The project mainly tracked SSIM, PSNR, and PCC.

5.1 SSIM

SSIM compares local image structure, contrast, and brightness. It was introduced by Wang, Bovik, Sheikh, and Simoncelli as a quality metric closer to human visual perception than simple error measures [R2]. In this project, SSIM is the most important number because virtual staining must preserve tissue structure.

5.2 PSNR

PSNR is based on mean squared error. It is useful for measuring pixel-level closeness. However, a model can have decent PSNR and still look biologically wrong if structures are shifted or blurred.

5.3 PCC

PCC measures correlation. In simple words, it checks whether bright and dark patterns rise and fall together between prediction and target. This is useful for biological texture consistency.

5.4 Why we did not trust one metric alone

No single metric is perfect. SSIM is important for structure, PSNR is useful for pixel accuracy, and PCC helps evaluate pattern correlation. The final default ensemble was chosen because it tied the best SSIM and improved PSNR/PCC:

Final Mode	SSIM	PSNR	PCC
0.30 v20 + 0.50 v22A + 0.20 v21B	0.7838	25.08	0.8789
0.20 v20 + 0.60 v22A + 0.20 v21B	0.7838	25.16	0.8794

Because both modes tied on SSIM, the second one became the default.

6. Timeline of Model Versions

The project did not jump directly to the final model. It went through many versions. Each version taught something.

Version	Main Idea	Result	Lesson
v1-v7	Pix2Pix on unregistered data	about 0.26 SSIM	Registration is required
v8-v10	Pix2Pix after TV-L1 registration	up to 0.706 SSIM	Registration unlocks learning
v11	Add VGG perceptual and Sobel losses	0.712 SSIM	Perceptual signal helps but has a ceiling
v12	Add HED stain loss	Failed	More losses can conflict
v13	Attention U-Net + more data	0.6326 SSIM	More data can hurt if quality is lower
v14	Simpler U-Net, curated patches	0.7080 SSIM	Simplicity and data quality help
v15	Attention U-Net on patches	0.7199 then diverged	Complex GAN training unstable
v16	Remove HED, full-size top-1k	0.6976 then diverged	Still unstable
v17	5-level U-Net + SSIM loss	0.379 or low ceiling	Wrong warm-start and loss choices can break learning
v19	DenseUNet + complex losses	Diverged	Complex loss stack not reliable
v19b	ResNet-34 U-Net + L1 only	0.7489 leaky split	L1-only is stable, but split was not clean
v20	ConvNeXt U-Net + L1	0.7549 leaky split	Strong encoder helps
v20_fixed	Clean split v20	0.7606 clean	Clean validation confirmed strength
v21A	Hibou-B frozen feature decoder	0.7605 clean	Histology encoder matched but did not beat v20
v22A	v20 warm-start on CLAHE content-quality data	0.7807 fixed CLAHE TTA	Better data improves final eval
v21B	Hibou-B on CLAHE content-quality data	0.7790 fixed CLAHE TTA	Useful ensemble complement
Final	Weighted v20/v22A/v21B TTA ensemble	0.7838 SSIM	Best overall pipeline

This timeline shows a clear pattern. Early progress came from registration. Middle progress came from stabilizing the learning setup. Final progress came from stronger encoders, better validation discipline, and improved registration/data selection.

7. Early Pix2Pix Experiments: Why They Were Not Enough

Pix2Pix is a famous image-to-image translation framework. It trains a generator to create an output image and a discriminator to judge whether the output looks real [R4]. This idea is natural for virtual staining because the input is one image type and the output is another image type.

The early v1-v7 experiments used a Pix2Pix-style U-Net generator and PatchGAN discriminator. The setup was reasonable, but the data was not aligned enough. Because of that, the model learned blurry averages instead of sharp biology.

This taught the first major lesson:

> A powerful image-to-image model cannot fix badly aligned supervision.

When the target image is shifted away from the input image, pixel losses and adversarial losses push the model in conflicting directions. The generator tries to satisfy impossible training examples, so the safest prediction becomes a smooth average.

8. v8-v10: Registration Unlocks the First Good Result

After TV-L1 registration, the Pix2Pix-style model improved sharply. v10 used a WGAN-GP style adversarial setup, which is a more stable version of Wasserstein GAN training that adds a gradient penalty [R6]. v10 reached about:

Metric	v10
SSIM	0.706
PSNR	22.75 dB

This was the first sign that the project was on the right track. The model architecture did not magically change the task; the data became learnable.

The team also used practical engineering improvements:

1. RAM caching so images did not need to be repeatedly loaded from disk.
2. Mixed precision training to use GPU memory more efficiently.
3. A ResNet-34 encoder to provide stronger image features.

ResNet introduced residual connections, which make deep networks easier to train [R7]. This was a good fit because histology images contain many repeated textures and edges.

9. v11: The First Strong Historical Baseline

v11 added VGG-19 perceptual loss and Sobel edge loss on top of the registered Pix2Pix setup. VGG networks are deep convolutional networks known for strong visual feature extraction [R8]. A perceptual loss compares feature maps inside a pretrained network instead of comparing only raw pixels.

Why this made sense:

1. H&E staining is not only about color. It is also about cell texture.
2. A raw pixel loss can punish tiny shifts too harshly.
3. A perceptual loss can compare larger visual patterns.
4. Sobel edge loss encourages structural boundaries.

v11 reached:

Metric	v11
SSIM	0.7120
PSNR	23.03 dB
PCC	0.8906

At that time, v11 was the best project result. It proved that registration plus meaningful feature loss could create realistic virtual staining.

However, v11 was not the final answer. The project later found that adversarial/perceptual complexity was not as reliable as a strong encoder plus clean L1 training.

10. Why Some "More Advanced" Ideas Failed

Several later models tried to improve performance by adding more components: HED stain loss, attention gates, multi-scale discriminators, MS-SSIM loss, and larger loss combinations.

Some of these ideas were scientifically reasonable. For example, HED decomposition tries to separate Hematoxylin and Eosin stain channels. Attention gates try to focus the model on important spatial regions. Multi-scale discriminators try to judge both local texture and global realism.

But in practice, many of these additions made training unstable.

The reason is that every loss function sends a gradient signal. If too many losses disagree, the model may not know what to optimize. One loss may push the output toward pixel accuracy, another toward texture realism, another toward edge sharpness, and another toward stain-channel separation. If the labels are not perfect, this can become worse.

The project learned this:

> A complicated training objective is not automatically better. If data alignment is imperfect, simple stable losses can win.

This is why v19b and v20 returned to L1-only training.

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11. v13-v16: Data Quantity Was Not the Same as Data Quality

One tempting idea was to train on more data. The project had thousands of registered pairs, so using all of them sounded useful. But v13 used many lower-quality registered pairs and dropped to about 0.6326 SSIM.

This was a major lesson:

> More data helps only if the labels are reliable enough.

In supervised learning, the model learns from the target image. If the target image is misaligned, the target becomes noisy. Adding more noisy pairs can make the model worse.

This is why the project later focused on top-K selection:

1. choose better registered pairs,
2. avoid negative-gain pairs,
3. prefer tissue-rich and information-rich patches,
4. evaluate using a fixed held-out set.

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12. v17 and v19: Why Warm-Starts and Losses Must Match the Architecture

v17 tried a deeper 5-level U-Net with L1 + SSIM loss. One attempt used an incompatible warm-start from a different architecture. That caused a severe failure, with SSIM around 0.379.

This is an important engineering lesson. Warm-starting means loading weights from an older model into a new model. It only helps if the old weights match the new architecture and scale. If they do not match, the model starts from a broken state.

v19 tried a stronger architecture with multiple losses but diverged. v19b simplified the setup to a ResNet-34 U-Net with L1 only. That model became stable and reached 0.7489, but later this was found to be a leaky split result. The model was useful as a sign that L1-only training worked, but it could not be used as the clean final claim.

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13. The Clean Split Problem

One of the most important scientific corrections in the project was identifying train/validation leakage.

In image patch projects, leakage can happen when patches from the same slide or same tissue region appear in both training and validation. The model may then be tested on data that is too similar to what it has already seen.

The project found that old v19b and old v20 had overlapping train/validation prefixes. Specifically, 91 out of 100 validation prefixes overlapped training in those older runs, according to the project research notes [P3].

This did not mean the models were useless. It meant their scores were not clean final validation scores.

v20_fixed corrected this by using a clean 900/100 split with `overlap=0`. This gave:

Model	Clean Split?	SSIM	PSNR	PCC
v20_fixed	Yes	0.7606	24.92	0.8652

This became the clean old-registered baseline.

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14. v20: Why ConvNeXt Became the Strongest Deployable Family

v20 used a ConvNeXt-Base encoder inside a U-Net-style image-to-image model. ConvNeXt modernized convolutional networks using design ideas learned from Vision Transformers while keeping CNN strengths [R5]. The exact deployed family used a `timm` ConvNeXt-Base CLIP/LAION pretrained checkpoint [R12].

Why ConvNeXt helped:

1. It is a stronger pretrained visual feature extractor than the earlier ResNet-34 setup.
2. It captures both local texture and broader image context.
3. It worked well with a simple L1 loss.
4. It avoided GAN instability.

The v20 lesson was:

> A strong pretrained encoder plus simple stable training beat a complicated GAN stack.

This became one of the main design principles for the final project.

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15. v21A and v21B: Why Hibou-B Was Tested

Hibou-B is a histopathology-focused foundation model available through Hugging Face [R11]. It is based on the idea that a model pretrained on histology images should understand tissue patterns better than a general natural-image model. This idea follows the broader trend of strong self-supervised visual representations such as DINOv2 [R10].

The team tested Hibou-B because it was a logical question:

> If the task is histology, can a histology-pretrained encoder beat a general visual encoder?

v21A used a frozen Hibou-B feature extractor with a decoder. It reached:

Model	SSIM	PSNR	PCC
v21A	0.7605	24.77	0.8634

This almost exactly matched v20_fixed but did not beat it as a single model.

Later v21B used Hibou-B with the content-quality CLAHE data. On the final CLAHE fixed evaluation it scored:

Model	SSIM	PSNR	PCC
v21B TTA4	0.7790	24.27	0.8598

v21B was not the best single model, but it added useful diversity to the final ensemble.

The lesson:

> Domain-specific pretraining helped as an ensemble member, but it did not replace the ConvNeXt v22A model.

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16. v22A: The Best Single Model on CLAHE Data

v22A was the most important final single-model experiment. It took the stable v20 architecture and warm-started it from `models/v20_fixed_model.pth`, then trained it on the content-quality CLAHE top-1000 dataset.

This design was conservative and smart:

1. Keep the architecture that already worked.
2. Keep the simple L1-based training behavior.
3. Improve the data instead of adding unstable losses.
4. Warm-start from a known good checkpoint.

On internal validation, v22A reached SSIM 0.7655. More importantly, on the final fixed CLAHE evaluation, v22A TTA4 reached:

Metric	v22A TTA4
SSIM	0.7807
PSNR	25.16 dB
PCC	0.8782

This showed that the CLAHE registration and content-quality training data produced real fixed-evaluation improvement.

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17. Why Test-Time Augmentation Was Used

Test-time augmentation is simple. The model predicts the output image multiple ways:

1. original image,
2. horizontally flipped image,
3. vertically flipped image,
4. both horizontally and vertically flipped image.

The flipped predictions are flipped back and averaged.

This helps because neural networks are not perfectly symmetric. If a small detail is predicted slightly differently after a flip, averaging can reduce noise.

In this project, TTA4 was used for the final comparison because it improved stability without retraining the model. The cost is slower inference. For the final app, this is acceptable in high-quality mode, but a faster single-pass mode can still be used if speed matters.

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18. Why the Final Ensemble Was Chosen

The final ensemble combines three model families:

1. v20_fixed: the clean old-registered ConvNeXt baseline.
2. v22A: the best single CLAHE-trained ConvNeXt model.
3. v21B: the Hibou-B histology-feature model.

The final tested ensembles were:

Ensemble	SSIM	PSNR	PCC
0.30 v20 + 0.50 v22A + 0.20 v21B	0.7838	25.08	0.8789
0.20 v20 + 0.60 v22A + 0.20 v21B	0.7838	25.16	0.8794

Both tied on SSIM. The second one had better PSNR and PCC, so it became the default.

Why the weights make sense:

Component	Weight	Reason
v20_fixed	0.20	Keeps robust old-distribution behavior
v22A	0.60	Best single CLAHE model, so it gets the largest weight
v21B	0.20	Adds histology-domain diversity

The ensemble improvement over v22A alone is small:

```
 v22A TTA4 SSIM: 0.7807
final balanced ensemble SSIM: 0.7838
gain: +0.0031 SSIM
```

This is not a huge jump, but at the final stage of image restoration tasks, small improvements can matter if they are consistent and evaluated fairly.

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19. Final Results Table

The final evaluation compared old registered data and CLAHE same-prefix data.

Fixed Evaluation Set	Best Method	SSIM	PSNR	PCC
Old registered	v20 TTA4	0.7634	25.03	0.8684
CLAHE same-prefixes	0.30 v20 + 0.50 v22A + 0.20 v21B TTA4	0.7838	25.08	0.8789
CLAHE same-prefixes	0.20 v20 + 0.60 v22A + 0.20 v21B TTA4	0.7838	25.16	0.8794

The project therefore has two honest deployment messages:

1. For the old registered input distribution, use v20 TTA4.
2. For the final CLAHE-trained pipeline, use the balanced v20/v22A/v21B ensemble.

This distinction matters. A model trained or tuned for one data distribution may not be best for another.

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20. The Final App

The project now includes a final app:

```
python best_stain_app.py
```

It also supports command-line inference:

```
python best_stain_app.py --input path/to/unstained.tif --output virtual_he.png
```

The app modes are:

Mode	Meaning
balanced	Default final ensemble
best_ssim	Tied SSIM ensemble with slightly lower PSNR/PCC
v22_tta	Best single CLAHE model
v20_tta	Best old-registered fallback

The final app needs these model files:

```
models/v20_fixed_model.pth
models/v22a_content_quality_top1000_ft_model.pth
models/v21b_hibou_content_quality_ft_model.pth
```

For the v21B model, the app also needs the Hibou-B Hugging Face model code/config. A clean laptop can use:

```
models/hibou-b/
```

or set:

```
HIBOU_MODEL_PATH
```

If Hibou-B is hard to distribute, the app can still run the v22A-only mode.

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21. Main Challenges and How They Were Solved

Challenge 1: Image pairs were not aligned

Problem: The same cell appeared in different positions in input and target.

Solution: Use TV-L1 optical flow registration, then improve it with CLAHE.

Why this decision was right: Registration gave the largest early improvement in the project, moving from about 0.26 SSIM to above 0.65 in the registered Pix2Pix line.

Challenge 2: More data sometimes made results worse

Problem: All 8,885 pairs included low-quality registrations.

Solution: Use top-K selection and content-quality ranking instead of blindly using everything.

Why this decision was right: v13 showed that using more low-quality data could reduce performance. v22A showed that better selected CLAHE data improved fixed evaluation.

Challenge 3: GAN training became unstable

Problem: Adding adversarial, perceptual, HED, and SSIM losses created conflicting gradients.

Solution: Move toward simple L1-only training with stronger pretrained encoders.

Why this decision was right: v19b/v20/v20_fixed became stable, and v22A produced the strongest single-model result.

Challenge 4: Validation leakage could mislead the team

Problem: Earlier v19b and v20 runs had overlapping train/validation prefixes.

Solution: Use a clean split for v20_fixed and a fixed final evaluation for v22A/v21B/final ensemble.

Why this decision was right: It made the final claim more trustworthy.

Challenge 5: Domain foundation models were harder to deploy

Problem: Hibou-B was useful but required external Hugging Face model files and custom loading.

Solution: Keep Hibou-B as an ensemble member but preserve v22A and v20 fallback modes.

Why this decision was right: The best internal model quality is available, but the app still has simpler distribution paths.

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22. Why the Final Decisions Are Defensible

The final system is defensible because each major choice has evidence.

Decision	Evidence
Use non-rigid registration	Unregistered Pix2Pix was about 0.26 SSIM; registered models exceeded 0.70
Use CLAHE TV-L1	All-pair registration mean SSIM improved from 0.4134 to 0.5045
Use clean validation	Earlier overlapping splits were corrected by v20_fixed and fixed eval
Use ConvNeXt U-Net	v20_fixed was the strongest clean old-registered baseline
Use v22A as main final model	v22A TTA4 was the best single CLAHE fixed-eval model
Include v21B in ensemble	It added diversity and improved final ensemble score
Make balanced ensemble default	It tied SSIM and improved PSNR/PCC
Keep v20 fallback	Old registered validation still preferred v20 TTA4

This is exactly how engineering research should work: test, measure, find failure modes, simplify when needed, and keep the evaluation fair.

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23. Limitations

The project is strong, but it is not finished clinical validation.

Limitation 1: Dataset and slide diversity

The final content-quality top-1000 dataset improved performance, but it may still be narrow. More slides and more tissue types are needed to test generalization.

Limitation 2: Metrics are not clinical proof

SSIM, PSNR, and PCC are useful, but they do not prove that a pathologist would make the same diagnosis from the virtual stain. A future study should include expert review and cell/nuclei-level quantitative checks.

Limitation 3: Final ensemble is slower

The default mode runs three models and four TTA passes. That can mean up to 12 model forward passes per image. It is high quality but slower than a single model.

Limitation 4: Hibou-B packaging

The v21B model requires Hibou-B runtime files. This makes full portable distribution harder. The app handles this by allowing v22A-only and v20-only fallback modes.

Limitation 5: The app is not a medical device

The app is a research prototype. It should not be used for clinical decisions without proper validation, regulatory review, and pathologist evaluation.

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24. Future Work

The next best steps are not random architecture changes. The project should continue the pattern that worked.

Step 1: Scale CLAHE/content-quality data carefully

Train on top-1500 and top-2000 selected pairs. Make sure slide diversity does not collapse.

Step 2: Use the same fixed evaluation gate

Every new model should be tested on the same fixed validation prefixes. Otherwise, scores will not be comparable.

Step 3: Add biological validation

Use nuclei segmentation, tissue-region analysis, and pathologist review. The final model should be judged not only by SSIM but by whether it preserves diagnostic structures.

Step 4: Try domain encoders again after data policy is stable

Hibou-B helped in the ensemble but did not win alone. A future domain encoder should be tested only after the data pipeline is stable.

Step 5: Improve public deployment

The best public package may be v22A-only because it avoids Hibou-B packaging complexity. The internal research package can keep the full ensemble.

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25. Final Takeaway

The project moved from blurry unregistered Pix2Pix outputs to a strong final ensemble by solving the right problems in the right order.

The biggest improvements were not from one magical model. They came from:

1. aligning the images,
2. selecting better training pairs,
3. using clean validation,
4. choosing stable training over complicated losses,
5. using a strong ConvNeXt backbone,
6. adding Hibou-B only where it helped,
7. combining models carefully at inference time.

The final result is:

```
 CLAHE TV-L1 + balanced v20/v22A/v21B TTA4 ensemble
SSIM 0.7838, PSNR 25.16 dB, PCC 0.8794
```

That is the current best project baseline.

Appendix A: Final App and Repository Files

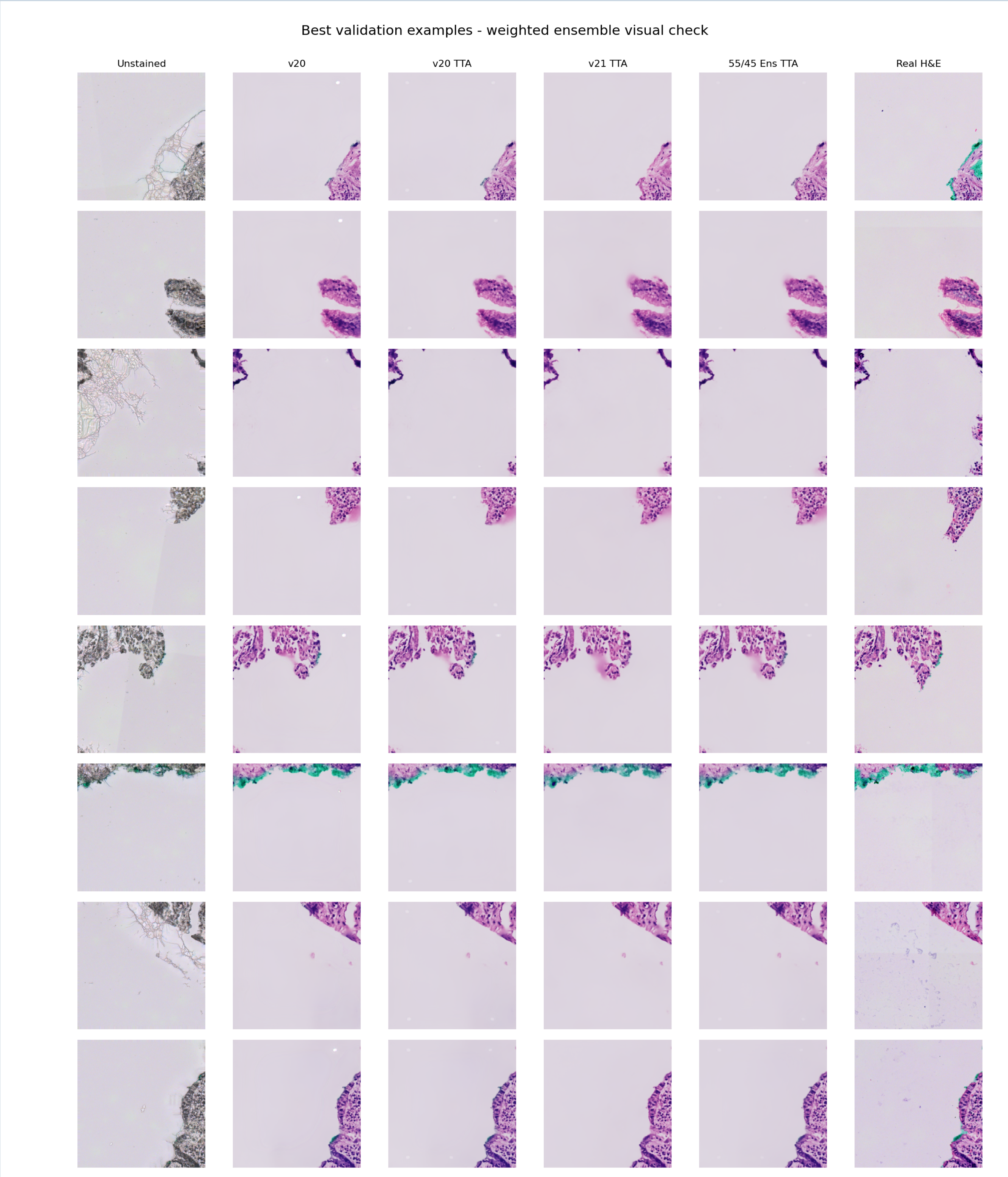
File	Purpose
`best_stain_app.py`	Final app
`best_model_manifest.json`	Final model manifest
`APP_DISTRIBUTION.md`	How to run/build/share the app
`SCRIPT_INDEX.md`	Root script handoff map
`logs/final_v21b_v22a_eval_summary.txt`	Final aggregate metrics
`logs/final_v21b_v22a_eval_per_pair.csv`	Per-pair final metrics
`registration_pipeline_clahe.py`	CLAHE TV-L1 registration
`train_v20_csv_variant.py`	v22A training path
`train_v21b_hibou_content.py`	v21B training path
`eval_v21b_v22a_final.py`	Final evaluation

Appendix B: Visual Example Sheet

The project includes generated visual grids under:

```
showcase_images/v21_ensemble_comparison/
```

The image below is included as a qualitative reference. Metric tables remain the main evidence.



Appendix C: Claim-Evidence Map

Claim	Evidence	Status
Registration was necessary	v1-v7 around 0.26 SSIM; registered v10 reached 0.706	Supported

Claim	Evidence	Status
CLAHE TV-L1 improved registration	all-pair mean SSIM 0.4134 -> 0.5045	Supported
v20_fixed is clean old-registered baseline	clean split, SSIM 0.7606	Supported
v22A is best single CLAHE model	fixed CLAHE TTA4 SSIM 0.7807	Supported
Final balanced ensemble is best handoff mode	SSIM 0.7838, PSNR 25.16, PCC 0.8794	Supported
Hibou-B helped but did not win alone	v21B TTA4 0.7790, below v22A 0.7807, but useful in ensemble	Supported
Clinical use still requires validation	Metrics are image similarity metrics, not pathologist diagnosis	Supported

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References

Project Artifacts

[P1] `logs/final_v21b_v22a_eval_summary.txt`, final v20/v22A/v21B fixed evaluation, generated in this repository. [P2] `best_model_manifest.json`, final weighted ensemble manifest, generated in this repository. [P3] `research/08_results_comparison.md`, project results matrix and leakage notes. [P4] `research/02_image_registration.md`, registration methods and CLAHE TV-L1 results. [P5] `research/04_training_history.md`, chronological model history from v1 to final ensemble.

External Technical References

[R1] Zach, C., Pock, T., and Bischof, H. "A Duality Based Approach for Realtime TV-L1 Optical Flow." DAGM 2007. https://link.springer.com/chapter/10.1007/978-3-540-74936-3_22

[R2] Wang, Z., Bovik, A. C., Sheikh, H. R., and Simoncelli, E. P. "Image Quality Assessment: From Error Visibility to Structural Similarity." IEEE Transactions on Image Processing, 2004. https://pubmed.ncbi.nlm.nih.gov/15376593/

[R3] Ronneberger, O., Fischer, P., and Brox, T. "U-Net: Convolutional Networks for Biomedical Image Segmentation." MICCAI 2015. https://arxiv.org/abs/1505.04597

[R4] Isola, P., Zhu, J.-Y., Zhou, T., and Efros, A. A. "Image-to-Image Translation with Conditional Adversarial Networks." CVPR 2017. https://arxiv.org/abs/1611.07004

[R5] Liu, Z. et al. "A ConvNet for the 2020s." CVPR 2022. https://arxiv.org/abs/2201.03545

[R6] Gulrajani, I. et al. "Improved Training of Wasserstein GANs." NeurIPS 2017. https://arxiv.org/abs/1704.00028

[R7] He, K., Zhang, X., Ren, S., and Sun, J. "Deep Residual Learning for Image Recognition." CVPR 2016. https://arxiv.org/abs/1512.03385

[R8] Simonyan, K. and Zisserman, A. "Very Deep Convolutional Networks for Large-Scale Image Recognition." ICLR 2015. https://arxiv.org/abs/1409.1556

[R9] Zuiderveld, K. "Contrast Limited Adaptive Histogram Equalization." Graphics Gems IV, 1994. https://dl.acm.org/doi/10.5555/180895.180940

[R10] Oquab, M. et al. "DINOv2: Learning Robust Visual Features without Supervision." 2023. https://arxiv.org/abs/2304.07193

[R11] Hugging Face model card for `histai/hibou-b`. https://huggingface.co/histai/hibou-b

[R12] Hugging Face model card for `timm/convnext_base.clip_laion2b_augreg_ft_in1k`. https://huggingface.co/timm/convnext_base.clip_laion2b_augreg_ft_in1k